Digital Signal Processing Laboratory

Analysis of Discrete-Time Signals and LTI Systems

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Abstract

This report outlines fundamental experiments in Digital Signal Processing (DSP), focusing on the properties and behaviors of discrete-time systems. The primary objectives include analyzing linearity, time-invariance, impulse and step responses, the mathematical equivalence of convolution and filtering, and Bounded-Input Bounded-Output (BIBO) stability.

1 Introduction

Discrete-time systems form the backbone of modern digital signal processing. Evaluating the characteristics of these systems—such as linearity and time-invariance—is crucial for predicting system responses to arbitrary inputs. This project utilizes MATLAB to mathematically and visually model various signal processing operations.

2 System Properties

2.1 Linearity and Non-Linear Systems

A system is considered linear if it satisfies the superposition principle (additivity and homogeneity). During the laboratory experiments, we observed that inserting an operation such as squaring the signal or introducing non-zero initial conditions violates linearity. For a basic nonlinear system, high-frequency inputs resulted in more rapid output variations compared to low-frequency inputs. Additionally, applying a DC shift (e.g., $x[n] = \sin(2\pi \cdot 0.05 \cdot n) + 10$) directly translates the amplitude of the signal by the corresponding constant.

2.2 Time-Invariance

A time-invariant system responds identically to a delayed input, shifting the output by the exact same delay. Experiments showed that while standard Linear Time-Invariant (LTI) difference equations maintain this property, the introduction of non-zero initial conditions causes the system to become time-variant, meaning the system's output depends on the absolute timeline of the input application.

3 System Responses: Impulse and Step

For an LTI system defined by the coefficient vectors `num = [0.9, -0.45, 0.35, 0.002]` and `den = [1, 0.71, -0.46, -0.62]`, the impulse response was evaluated using two distinct MATLAB functions:

1. Utilizing the built-in `impz` function.
2. Utilizing the `filter` function by generating an explicit Kronecker delta input (a single 1 followed by zeros).

Both methodologies yielded identical output arrays, validating the internal mechanisms of the functions. Replacing the impulse input with an array of ones (`ones(1, N)`) successfully generated the system's step response.

4 Convolution versus Filtering

To demonstrate the functional equivalence of time-domain convolution and filtering for finite sequences, we processed an input sequence $x[n]$ with an impulse response $h[n]$. Using the `conv` function, the resulting sequence length is inherently $L_x + L_h - 1$[cite: 1]. To replicate this exact behavior using the `filter` function without truncating the output, we padded the input sequence $x[n]$ with a quantity of zeros equal to $L_h - 1$.

$$y_{conv}[n] = x[n] * h[n] \equiv y_{filter}[n] = \text{filter}(h, 1, [x, \text{zeros}]) \tag{1}$$

The MATLAB subplots confirmed that zero-padding effectively aligns the discrete filtering process with standard linear convolution.

5 BIBO Stability Analysis

A discrete-time LTI system is Bounded-Input Bounded-Output (BIBO) stable if and only if its impulse response is absolutely summable:

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty \tag{2}$$

An experiment was conducted on the system described by the difference equation:

$$y[n] + 1.5y[n-1] + 0.9y[n-2] = x[n] - 0.8x[n-1] \tag{3}$$

A `for`-loop was constructed to iteratively sum the absolute values of the impulse response, halting if the absolute value fell below a predefined threshold of 10^{-6}. The execution revealed that the sum did not converge, and the impulse response exhibited unbounded, oscillatory behavior. It was concluded that the system is unstable because its poles are located outside the unit circle in the z-plane.

6 Conclusion

Through computational modeling in MATLAB, the foundational properties of discrete-time DSP systems were successfully verified. LTI assumptions require strict zero-initial conditions, while convolution can be accurately replicated via filtering mechanisms by employing zero-padding. Furthermore, iterative algorithmic approaches effectively demonstrated system instability through the divergence of the impulse response.